

CMR Scaling Law: Predicting Critical Mixture Ratios for Continual Pre-training of Language Models

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Abstract

Large Language Models (LLMs) excel in diverse tasks but often underperform in specialized fields due to limited domain-specific or proprietary corpus. Continual pre-training (CPT) enhances LLM capabilities by imbuing new domain-specific or proprietary knowledge while replaying general corpus to prevent catastrophic forgetting. The data mixture ratio of general corpus and domain-specific corpus, however, has been chosen heuristically, leading to sub-optimal training efficiency in practice. In this context, we attempt to re-visit the scaling behavior of LLMs under the hood of CPT, and discover a power-law relationship between loss, mixture ratio, and training tokens scale. We formalize the trade-off between general and domain-specific capabilities, leading to a well-defined Critical Mixture Ratio (CMR) of general and domain data. By striking the balance, CMR maintains the model’s general ability and achieves the desired domain transfer, ensuring the highest utilization of available resources. Therefore, if we value the balance between efficiency and effectiveness, CMR can be consider as the optimal mixture ratio. Through extensive experiments, we ascertain the predictability of CMR, and propose CMR scaling law and have substantiated its generalization. These findings offer practical guidelines for optimizing LLM training in specialized domains, ensuring both general and domain-specific performance while efficiently managing training resources.

1 Introduction

Large Language Models (LLMs) exhibit versatile abilities, including question answering, translation, summarization, role-playing, etc. (Brown et al., 2020; Touvron et al., 2023a,b; Li et al., 2023). Their performance, however, may degrade in specific domains due to limited corresponding pre-training data. To enhance LLMs’ abilities in specialized areas and avoid the enormous cost of

re-training, a popular approach is Continual Pre-Training (CPT) (Colombo et al., 2024; Chen et al., 2023; Yıldız et al., 2024; Luo et al., 2023). This approaches are likely to equip LLMs with new domain-related capabilities without much general performance penalty.

Although CPT has been proven effective on multiple domains such as code (Li et al., 2023; Lei et al., 2024), law (Colombo et al., 2024) and medicine (Chen et al., 2023), the interplay among loss prediction and its scaling behavior with model size, and the number of training tokens is yet to be fully explored. Additionally, the composition of continual pre-training data is simply set up in a heuristic manner (Colombo et al., 2024; Chen et al., 2023), far from being principled. An inappropriate mixture ratio can lead to inefficient training (requiring excessive computation to adapt to specific domains) or insufficient training (failing to adequately reduce domain-specific loss). In light of this, three question hurdles we need to cross are as follows:

Does the optimal data mixture ratio exist for CPT? If so, how does it evolve with model scale or training token volume? Are there any involved simple yet principled laws?

Currently, several studies examine the scaling laws associated with different data mixture ratios. For instance, Ye et al. (2024) investigate how data mixtures shape scaling laws in the pre-training phase from the ground up, while Que et al. (2024) seek to pinpoint the optimal data mixture ratio in CPT, but overlook its crucial connection with the essential trade-off between general and domain loss in CPT.

Therefore, to strengthen our understanding about CPT and guide the experiments in the future, we attempt to address these questions with empirical studies on CPT of LLMs. Specifically, we pre-train several LLMs with different model sizes from scratch and perform CPT on downstream domains

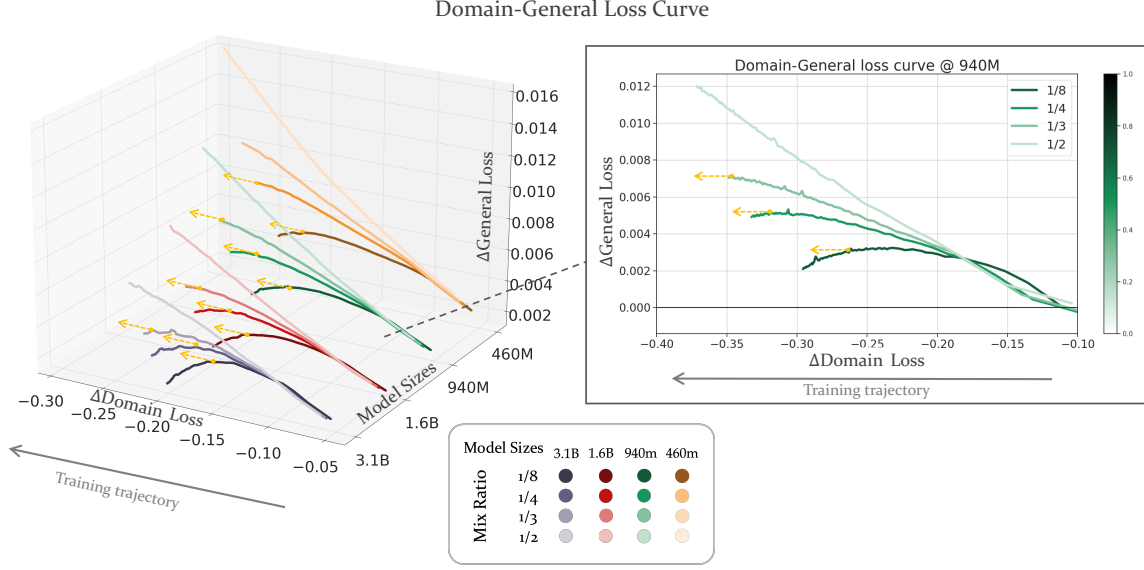


Figure 1: Follow the direction of the training trajectory to track the trend of the curve. Each bunch of lines represents a model size scale: {3.1B, 1.6B, 940M, 460M} and each group of line colors represents the mixture ratios {1/8, 1/4, 1/3, 1/2} from dark to light. In order to better display the trend, we have omitted proportions greater than 1/2. The yellow dashed lines $\leftarrow$ - point horizontally, indicating the corresponding ratios where $d\mathcal{L}_{\Delta\text{gen}}/d\mathcal{L}_{\Delta\text{dom}}$ closed to 0. The third set of lines of model size 940M, which has been zoomed in and depicted on the right side, showing the trend of the training curve more apparently. All horizontal and vertical cross-sections of the 3D diagram on the left side are detailed in the Appendix D.

(Finance and Academic Papers) with different data-mixture ratios. Our main contributions can be summarized as follows:

Formalization of the Trade-Off in CPT. We formalize the balance between domain-specific and general abilities during CPT by introducing the concept of feasible mixture ratios. CPT under feasible mixture ratios maintains performance on general data while enhancing performance on domain-specific data. We identify the maximum feasible mixture ratio as the **Critical Mixture Ratio (CMR)**, and regard it as the optimal mixture ratio by our definition.

Predictability of CMR. Through extensive experiments, we identify a power-law relationship between loss and both data-mixture ratio and training tokens. As such, we propose CMR scaling law to predict the best mixture ratio by scaling training token volume, which is verified to be generalizable.

Significance of CMR Scaling Law. CMR scaling law for CPT is crucial for efficient domain transfer for LLMs. This law allows us to determine the most efficient training configuration by predicting CMR using limited data and compute resources. Our finding provides valuable insights into the dynamics of CPT and offers practical guidelines for

optimizing LLM training in specialized domains.

2 Key Results

We train a series of LLMs with multiple mixture ratios of domain-specific data and general data to analyse the scaling behaviour in CPT. The method is detailed in section 3.2. Overall, we have the following key results:

1. The trade-off between two goals of CPT (Definition 1) suggests that, given a model of certain size, there exists a set of feasible mixture ratios (Definition 2) that achieve the goals under specific training data constraints.
2. Basically, general losses in CPT increase initially before decreasing, whereas domain losses tend to decrease. The relationships between loss and mixture ratio, as well as training volume, fit well with a power-law form, allowing for loss prediction under different mixture ratios and training tokens.
3. Using the loss prediction by mixture ratio and training volume, we can predict the CMR (Definition 3) with CMR scaling law. Given the maximum amount of training tokens, experiments in Figure 1 and predicted results in

Figure 5 both show that CMR goes up with increasing model scale: from 29.76% for the 460M model to 34.89% for the 940M model.

3 Background and Methods

The scaling law in the pre-training stage has been widely studied. In this work, we simplify the form of scaling law as much as possible, which is essentially consistent with previous works in Section 6.

In this section, we will elaborate on the three main concepts involved in this work, including objective of CPT (Definition 1), feasible mixture ratio (Definition 2) and CMR (Definition 3). Then, we describe our experiment setups, including data preparation, experiment procedures and evaluation.

3.1 Continual Pre-training on Mixed Dataset

Definition 1. Objective of CPT

Given the pre-trained LLM M_S of model size S , general dataset $\mathcal{D}_{\text{gen}}$, and domain-specific dataset $\mathcal{D}_{\text{dom}}$, we continually pre-train M_S on a mixed dataset $\mathcal{D}_R$, where the **mixture ratio** of the domain-specific data is R , with $R \in [0, 1]$. The mixed dataset $\mathcal{D}_R$ is denoted as $\mathcal{D}_R = \mathcal{D}_{\text{dom}} + \mathcal{D}_{\text{gen}}$ and $R = \frac{|\mathcal{D}_{\text{dom}}|}{(|\mathcal{D}_{\text{gen}}| + |\mathcal{D}_{\text{dom}}|)}$. We define $\mathcal{L}_{\text{gen/dom}}(M_S)$ as the domain or general loss of the model M_S . We denote $\mathcal{L}_{\text{gen/dom}}^{\text{CPT}}(M_S, \mathcal{D}_R, T)$ as the domain/general loss of model M_S after CPT on dataset $\mathcal{D}_R$ with training token volume T . The goals for CPT are formalized as follows:

1. By the end of training, the general loss is supposed to either reach plateau or head downward (within a certain tolerance $\epsilon \geq 0$):

$$\mathcal{L}_{\text{gen}}^{\text{CPT}}(M_S, \mathcal{D}_R, T_{\text{max}}) \leq \mathcal{L}_{\text{gen}}(M_S) + \epsilon. \quad (1)$$

2. Domain-specific loss should decline largely:

$$\mathcal{L}_{\text{dom}}^{\text{CPT}}(M_S, \mathcal{D}_R, T_{\text{max}}) < \mathcal{L}_{\text{dom}}(M_S). \quad (2)$$

The increase in T from 0 to T_{max} corresponds to the progression of the training trajectory. To better integrate these two aspects, we adopt the method of Lagrange multipliers (Rockafellar, 1993). The loss function $F(\cdot)$ for the whole objective of CPT is the Lagrangian as follows:

$$\begin{aligned} F(S, R, T, \lambda) = & \mathcal{L}_{\text{dom}}^{\text{CPT}}(M_S, \mathcal{D}_R, T) \\ & + \lambda(\mathcal{L}_{\text{gen}}^{\text{CPT}}(M_S, \mathcal{D}_R, T) \\ & - \mathcal{L}_{\text{gen}}(M_S) - \epsilon), \end{aligned} \quad (3)$$

where λ is the Lagrange multiplier used to enforce the constraint on the general loss while minimizing the domain-specific loss. In practice, λ governs the importance of two target dimensions in CPT. $F(S, R, T, \lambda)$ is the whole objective function.

Under resource constraints, the optimal training configuration should minimize $\mathcal{L}_{\text{dom}}$ while satisfying the constraint on $\mathcal{L}_{\text{gen}}$, which involves finding the optimal S , R , and T by solving the following optimization problem:

$$\begin{aligned} S^*, R^*, T^* = & \operatorname{argmin}_{M_S, R, T} F(S, R, T, \lambda), \\ \text{s.t. } & \begin{cases} \mathcal{L}_{\text{gen}}^{\text{CPT}}(M_S, R, T_{\text{max}}) \leq \mathcal{L}_{\text{gen}}(M_S) + \epsilon, \\ R \geq 0, T \geq 0, \lambda \geq 0. \end{cases} \end{aligned} \quad (4)$$

Definition 2. Feasible Mixture Ratio

Given fixed model size S , the optimization problem in Equation (3) can be boiled down to $F(R, T, \lambda)$. We first introduce a mixture ratio set $\mathbb{A}$: according to the first constraint of Definition 1, under a certain tolerance ϵ for the deterioration in the final general performance, we can choose an set of mixture ratios $\mathbb{A}$ satisfying $\mathbb{A} = \{R \mid \mathcal{L}_{\text{gen}}^{\text{CPT}}(M_S, R, T_{\text{max}}) \leq \mathcal{L}_{\text{gen}}(M_S) + \epsilon\}$. Ratios in $\mathbb{A}$ that align with our CPT objective are considered as feasible mixture ratios, denoted as the set $\mathbb{F}$. A detailed definition transformation is presented in Appendix B.2, and here we directly provide the formula and the results of derivation: within the feasible mixture ratios, there exists a point T_0 over the training trajectory of CPT. As CPT proceeds with $T > T_0$, we have $\mathbb{F} = \{R \mid \exists T_0 \in (0, T_{\text{max}}) : \frac{\partial F}{\partial T} \leq 0, R \in \mathbb{A}\}$. An equivalent condition of defining $\mathbb{F}$ can be derived as:

$$\begin{aligned} \mathbb{F} = & \{R \mid \exists T_0 \in [0, T_{\text{max}}] \\ & : \left. \frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)} \right|_R = -\frac{1}{\lambda} < 0, R \in \mathbb{A}\}. \end{aligned} \quad (5)$$

For simplicity, we have defined $\mathcal{L}_{\Delta \text{dom}} = \mathcal{L}_{\text{dom}}^{\text{CPT}} - \mathcal{L}_{\text{dom}}$ and $\mathcal{L}_{\Delta \text{gen}} = \mathcal{L}_{\text{gen}}^{\text{CPT}} - \mathcal{L}_{\text{gen}}$.

Visualization As shown in Figure 1, the training curves meeting the objective of CPT are marked with yellow dotted arrows, indicating the curves show a downward trend as training proceeds. The domain loss continuously decreases ($\mathcal{L}_{\Delta \text{dom}} \downarrow$) and the general loss is bounded ($d\mathcal{L}_{\Delta \text{gen}}/d\mathcal{L}_{\Delta \text{dom}} \rightarrow 0$) along the training trajectory until the ends of training. This visual representation effectively illustrates the behavior described by Equation (5),

demonstrating the trade-off relationship between the domain loss and the general loss during training. The specific derivation and the interpretation of the slope for Figure 1 is detailed in Appendix B.2.

Definition 3. Critical Mixture Ratio (CMR)

Given limited compute resources and fixed model size, we hope that the language model can digest domain knowledge more efficiently by achieving the objective as described in Definition 1. Therefore, we define the maximum among feasible mixture ratios as the Critical Mixture Ratio (CMR) $R^* = \max\{R | R \in \mathbb{F}\}$.

The rationale is straightforward: if the ratio is less than CMR, the domain data is not sufficiently utilized in CPT; otherwise, the expected objective can’t be achieved, which is manifested as a intolerable increase in general loss, leading to degradation in general ability. Thus, we argue that the CMR is the most suitable ratio for CPT due to the ideal balance of two sides.

3.2 Method

Data preparation Our general pre-training data is composed of corpora from Chinese, English, and code. The Chinese corpus and English corpus both include articles from encyclopedia, books, news, papers and social media sites. The code corpus is a subset sampled from StarCoder (Li et al., 2023). The general pre-training dataset comprises a total of 220 billion tokens. The proportions of Chinese, English, and code are roughly 44% : 36% : 20%.

We meticulously craft two specific domain datasets for CPT: *Finance* and *Academic Papers*. The Finance dataset include financial news, financial policies and regulations, company announcements and research reports from securities and fund companies. The Academic Papers exclusively include papers from Arxiv. Each of the datasets contains at least 20 billion tokens, which is sufficient for our CPT.

Unless stated explicitly, all the following results are based on experiments with *Finance*. The results of CPT on *Academic Papers* are reported in Section 5.3.

LLM Architecture The involved LLMs in this study have the same architecture as Llama series (Touvron et al., 2023a,b) with standard multi-head attention. The number of parameters ranges from 460M to 3.1B. The architecture is detailed in Table 1 of Appendix.

Experiment Setup We split the general pre-training dataset into two subsets: a 200B-token general dataset for general pre-training and a 20B-token general dataset for CPT.

In the pre-training stage, we pre-train the LLMs from scratch with 200B-token general dataset with a max learning rate of $3e-4$, a batch size of 512, and a sequence length of 4096. The training step is 100,000 for each LLM. In the CPT stage, we train each LLM for another 10,000 steps (20 billion tokens) with a max learning rate of $3e-5$ and warmup-constant LR schedule, on a mixture of the 20B-token general dataset and a domain dataset with different mixture ratios.

Evaluation Scaling laws emphasize the predictability of pre-training loss (Kaplan et al., 2020; Hoffmann et al., 2022; Gao et al., 2023; Hernandez et al., 2021), which is a widely-used performance indicator. Recent studies (Du et al., 2024; Yuan et al., 2023) highlight that pre-training loss is highly correlated with downstream task performance. Therefore, we use the pre-training loss on the validation set to measure the model’s capability of general or domain-specific task during the CPT process. In addition, we use Mean Squared Error (MSE) and R-square (R^2) to measure the quality of the fitting, which provides a clear and interpretable analysis of the errors.

4 Does the Critical Mixture Ratio Exist?

— Yes, the CMR does exist.

A larger mixture ratio implies a higher proportion of domain-specific data in the training set, resulting in a lower domain loss. However, due to the potential catastrophic forgetting of domain transfer, it is essential to ensure that the loss in the new domain continues to decrease while the original capabilities of LLMs are preserved and not compromised during CPT. Consequently, a higher mixture ratio is not always best. This raises an important question: does a Critical Mixture Ratio (CMR) exist that can balance these two goals of CPT in Definition 1 effectively and efficiently?

Figure 1 (left) demonstrates that for models of various sizes, there is at least one curve at a specific ratio that shows a downward trend, highlighted by yellow dotted arrows. This indicates the presence of feasible mixture ratios that align with our CPT objective. On the other hand, larger models tend to have bigger feasible mixture ratios set $\mathbb{F}$ (more curves with yellow dotted arrows). For curves that

meet the objective of CPT, a higher ratio is preferable, as it incorporates more domain knowledge while optimizing training efficiency within the tolerance of decline in general capacity. Therefore, the critical mixture ratio is defined as the highest proportion among these satisfactory curves, representing the optimal ratio for the given model size and limited training token volume.

If feasible ratios exist, we can conclude that CMR is also supposed to exist. Fundamentally, the existence of CMR arises from the trade-off between general and domain-specific capabilities, as well as the limited data and computing resources. According to definition 3, the CMR is present across models of different scales, as shown in Figure 1. This figure illustrates the existence of CMR as the maximum value within the feasible set. However, the precise value of CMR can not be determined from the figure, as it requires extensive experiments with different mixture ratios. The estimation of CMRs is discussed in 5.3 and plotted in Figure 5.

To look closely, we enlarged the longitudinal section of M_{940M} in the 3D graph in Figure 1 and placed it on the right side. It can be seen that as the mixture ratio increases, the curve continues to rise until the loss in the general domain exceeds our tolerance. The potentially controversial issue is that the downward trend in one-third of the curves is as clear as in the rest. The reason why it is feasible curve here can be found in Appendix B.2. Although it is not easily noticeable, there are indeed points on this curve where the slope is less than 0.

Findings From another perspective, we plot the loss curves of the models under the same mixture ratio as shown in Figure 2. When the mixture ratio is $1/4$, all models can achieve the training objective of CPT. However, at a $1/3$ mixture ratio, only M_{940M} , $M_{1.6B}$ and $M_{3.1B}$ achieve the CPT goal. This indicates that CMR for M_{940M} is around $1/4$ within the scope of our training token volumes, while the CMRs for M_{940M} , $M_{1.6B}$ and $M_{3.1B}$ are at least $1/3$. In other words, **CMRs slightly increase with model size, suggesting that larger models can accommodate a higher proportion of domain data.** We also further this finding by taking more cross-sections of Figure 1 (left) in Appendix D and the predicted CMR in following section 5.

This phenomenon can be explained by the models' ability to consume domain knowledge. As the proportion of domain-specific data increases, the

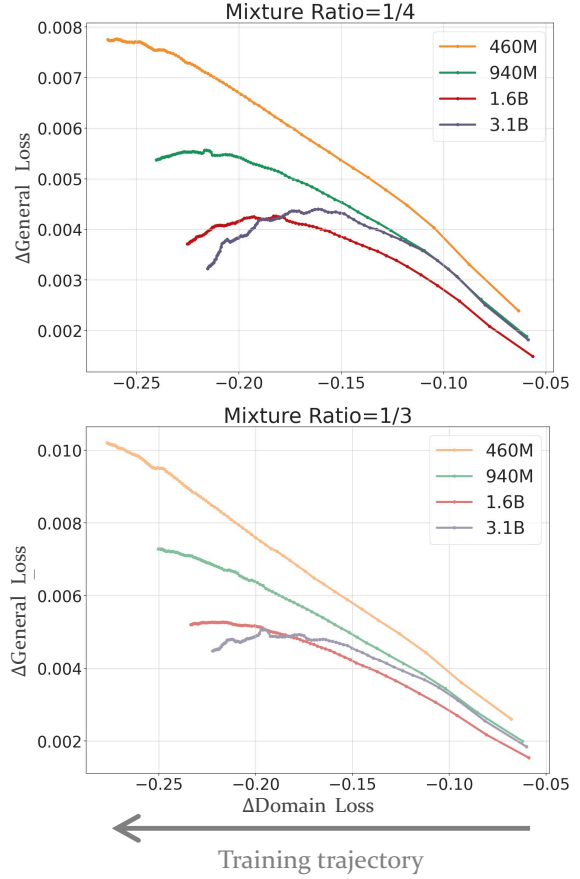


Figure 2: Follow the direction of the training trajectory to track the trend of the curve. The $\mathcal{L}_{\Delta_{\text{gen}}}$ and $\mathcal{L}_{\Delta_{\text{dom}}}$ loss functions for the models at mixture ratios of $1/4$ and $1/3$ are illustrated.

knowledge that the model needs to learn also increases. LLMs with smaller size struggle to absorb much of domain knowledge while preserving the general knowledge, leading to a degradation in their original general performance. In contrast, models with larger sizes can accommodate more knowledge with more parameters, thereby maintaining better performance.

5 Is CMR Predictable?

— Yes, the CMR can be predicted.

The existence of CMR indicates that in the process of CPT, we may explore the CMR scaling law to seek the best mixture ratio under resource constraints and domain data limitations, thereby optimizing training effectiveness and efficiency. In other words, the next question to answer is whether we can predict the CMR for model M_s given a maximum amount of continuation training token volume, T_{max} .

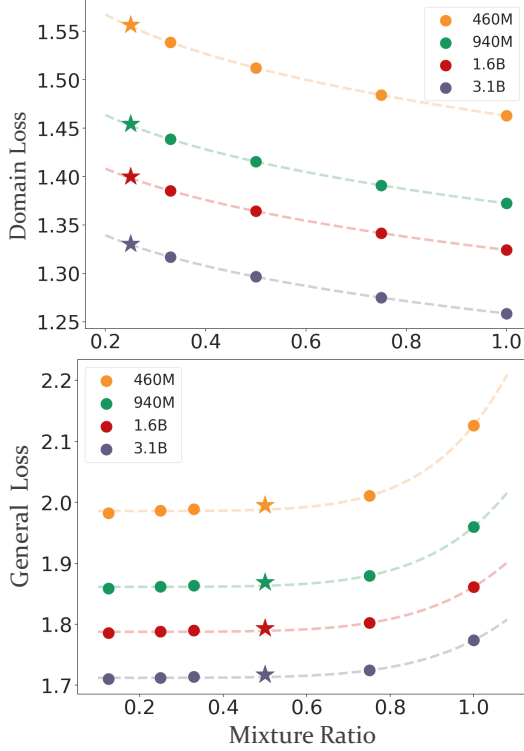


Figure 3: The upper figure shows the fitting curve of domain loss $\mathcal{L}_{\text{dom}}$ with the change of mixture ratio R , and the lower figure shows the fitting curve of general loss $\mathcal{L}_{\text{gen}}$. The solid circles (●) represent real losses, and the stars (★) represent the predicted losses.

To this end, two basic prerequisites must be met: predicting losses for different mixture ratio and predicting losses for different training token volume. In this section, we will demonstrate that these two prerequisites have been satisfied separately in 5.1 and 5.2, and finally detail the scaling law to predict CMR in 5.3. To keep notations simple, we omit fixed variables in the loss function ($\mathcal{L}_{\text{dom/gen}}$ and $\mathcal{L}_{\Delta\text{dom}/\Delta\text{gen}}$) in this section.

5.1 Predicting Losses of Mixture Ratio

Predicting the general and domain loss is closely related to understanding the scaling behavior in the CPT stage. We study the scaling behavior of losses at $T = T_{\text{max}}$. In addition, since scaling law aims to fit data points, their parametric forms should be intrinsically related to the observed trends in the data points. Based on previous works (Kaplan et al., 2020; Hoffmann et al., 2022) and data trends we observed, we proposed the simplified expression $\mathcal{L}(R)$ as a power-law form of

$$\mathcal{L}(R) = \alpha \cdot R^s + \beta,$$

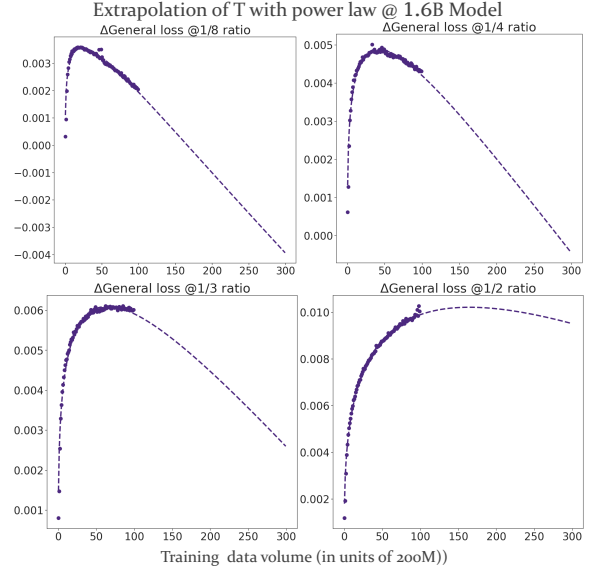


Figure 4: The figure shows the general loss of $M_{1.6B}$ fitting and extrapolating at four distinct mixture ratios: $\{1/8, 1/4, 1/3, 1/2\}$. As the ratio increases, the curve gradually rises when training data volume increases.

where α is a coefficient, s is the exponent, and β is the bias.

As shown in Figure 3, domain loss gradually decreases with the increase of the mixture ratio, while general loss remains almost unchanged initially and then begins to rise. After fitting the general loss and domain loss separately for different mixture ratios R (non-endpoint values, $R \in (0, 1)$), we make predictions on new ratios. As shown in Figure 3, the predicted values align closely with the fitted curve. Notably, the predictions demonstrate high accuracy, with error values within 0.05% as presented in Table 2.

Given the predicted $\mathcal{L}_{\text{dom}}(R)$ and $\mathcal{L}_{\text{gen}}(R)$ under different mixture ratios, we can obtain a range of mixture ratios that fulfil the tolerance limit ϵ , denoted as $\mathbb{A}$, according to Equation 3. In the objective of CPT we set, $\epsilon = 0.05$.

5.2 Predicting Losses of Training Tokens

Previous works (Kaplan et al., 2020; Hoffmann et al., 2022) have shown that the model size S and the volume of training tokens T can be used to fit the power law of loss. However, our work differs in two key aspects. First, we model the change of loss $\mathcal{L}_{\Delta\text{dom}/\Delta\text{gen}}(T)$ rather than the loss itself. Second, due to the phenomenon of general loss initially increasing and then decreasing as shown in Figure 4, we leverage a two-term polynomial function for better fitting. According to Equation 9

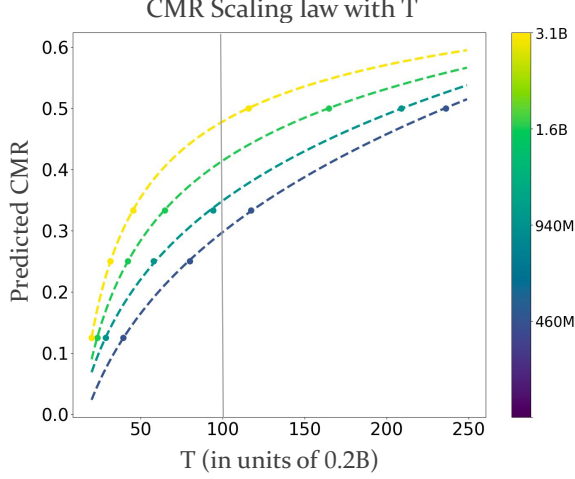


Figure 5: We can use the CMR scaling laws to predict CMRs under fixed model size S , and are extrapolated to $T = 250$, which is equivalent to a training volume of 500B tokens.

in Appendix B.2, the loss for CPT training tokens T is formulated as follows:

$$\begin{cases} \mathcal{L}_{\Delta\text{dom}}(T) = \alpha_1 \cdot T^{s_1} + \beta_1, \\ \mathcal{L}_{\Delta\text{gen}}(T) = \alpha_2 \cdot T^{s_2} + \alpha_3 \cdot T^{s_3} + \beta_2. \end{cases} \quad (6)$$

where $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, s_1, s_2$, and s_3 are learnable parameters. Our results demonstrate that the form (6) exhibits high fitting accuracy with low MSE and high R^2 in Table 3 and Figure 4.

5.3 Predicting CMR

According to the definition of feasible mixture ratios in Definition 2 and the method for determining the set $\mathbb{F}$ in Appendix B.2, where $\mathbb{F} \subset \mathbb{A}$, and $\mathbb{A}$ is obtained by predicting losses for any mixture ratio in 5.1, we can establish a relationship between training token volume T and the feasible mixture ratios by the fitting laws in 5.2. Overall, based on the parameters provided in Formula 6, the critical solution T_0 is obtained for a specific mixture ratio R_0 denoted as (derivation detailed in Appendix B.2):

$$T_0 = \left[-\frac{\alpha_1 \cdot s_1}{\lambda \alpha_2 \cdot s_2} \left(1 + \frac{\alpha_3 \cdot s_3}{\alpha_2 \cdot s_2} T_0^{s_3-s_2} \right)^{-1} \right]^{\frac{1}{s_2-s_1}} \quad (7)$$

When T_0 is less than the given maximum training token volume $T_{\max}$, we can conclude that the current ratio R_0 is a feasible mixture ratio. Conversely,

if T_0 exceeds $T_{\max}$, then R_0 is not a feasible mixture ratio. If T_0 is equal to $T_{\max}$, then R_0 is the critical ratio. We propose the following CMR scaling law:

$$R_{\text{CMR}} = \alpha_4 \cdot T^{s_4} + \beta_3. \quad (8)$$

The fitting curves are showed in Figure 5. In our experiments, $T_{\max}$ is 20B tokens, which corresponds to a value of $T = 100$ in the figure. Therefore, for four models of different scales, their predicted CMR are 29.8%, 34.9%, 41.4% and 47.8% for $M_{460\text{M}}, M_{940\text{M}}, M_{1.6\text{B}}, M_{3.1\text{B}}$, respectively.

Generalization In order to verify whether the CMR scaling law can be generalized, we experiment on another domain *Academic Papers* with different mixture ratios. In this generalization experiment, we only conduct CPT on the 460M-sized model with *Academic Papers* data proportions set to $\{1/8, 1/4, 1/2, 3/4, 1/3\}$ respectively. All other settings were kept consistent with *Finance*. As shown in Figure 6, the trade-off of CPT still exist in this domain, and thus there exists a CMR. Furthermore, the CMR scaling law still work, which can observed in Figure 7. The predicted CMR for *Academic Papers* is 36.7%, given the maximum training token volume $T_{\max} = 100$.

Open Discussion As showed in Figs. 5 and 7, the larger the $T_{\max}$, the wider the range of feasible mixture ratios. Therefore, it seems that when $T_{\max}$ tends to be infinity (the amount of data available for continued training is infinite, and computational resources are unlimited completely), the range of feasible mixture ratios would approach $(0, 1)$, leading CMR approaching 1. In this sense, each curve of CPT trajectory will show an expected convergence trend of objective, provided that there is enough T to allow it to develop.

Moreover, we find out that the solution of T_0 in Definition 2 approaches the inflexion point of $\mathcal{L}_{\Delta\text{gen}}(T)$ in Figure 4 and Figure 6, when $\lambda \rightarrow +\infty$, which we used for solving equations 14 ranges from 100 to 7000. The reason is likely to be that, the change in the general loss is much smaller than the change in domain, and λ in the objective function of CPT needs to be very large to amplify such subtle changes within the tolerance of constraints. In addition, during the training process, the decreasing trend of domain loss has always been present, but there are obvious inflection points in the of general loss curve (rise first and then fall).

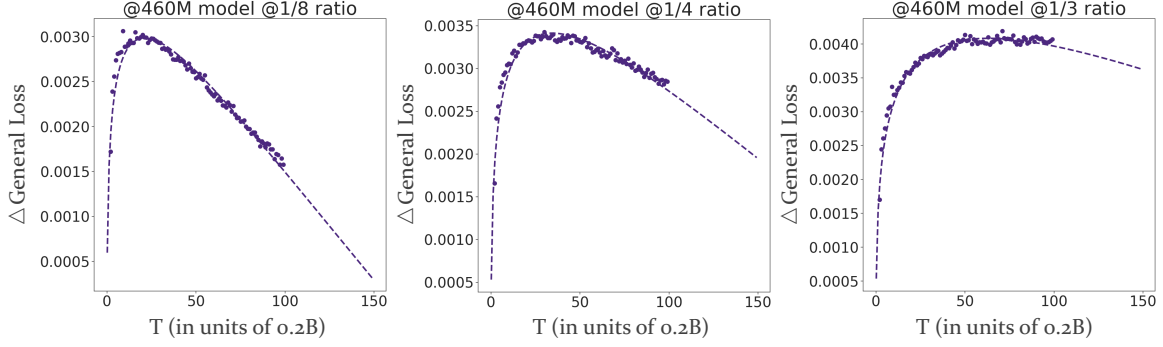


Figure 6: The figure shows the general loss of M_{460M} fitting and extrapolating at three distinct mixture ratios: $\{1/8, 1/4, 1/3\}$ with CPT on *Academic Papers*.

That is to say, by only locating the inflection points on the general loss curve and finding this distance to the max training token volume, we can roughly estimate how far away we are from CMR at current ratio.

6 Related Work

6.1 Continual Pre-training

Continual Pre-Training (CPT) aims to perpetually pre-train large language models (LLMs), allowing them to adapt to new domains and reducing the high costs associated with training models from scratch for specialized tasks (Yildiz et al., 2024). CPT can be employed to tailor LLMs for specific fields, such as code (Lei et al., 2024; Li et al., 2023), medicine (Chen et al., 2023), law (Colombo et al., 2024), and science. By using an appropriate mixture of data from various domains (Gururangan et al., 2020), CPT not only enhances downstream performance but also mitigates the issue of catastrophic forgetting, which is prevalent in all forms of post-training (Cossu et al., 2022; Luo et al., 2023).

6.2 Scaling Law

Numerous studies (Hestness et al., 2017; Henighan et al., 2020; Bahri et al., 2021; Kaplan et al., 2020; Hoffmann et al., 2022; Yao and Wang, 2023) demonstrate a power-law relationship between performance and the increase in both the number of parameters and the size of the training data. These relationships are crucial for large language models (LLMs), being of paramount importance in various stages such as pre-training (Kaplan et al., 2020; Hoffmann et al., 2022; Ye et al., 2024), supervised fine-tuning (SFT) (Hernandez et al., 2021; Lin et al., 2024), etc. Recently, researchers describe scaling laws from various different perspectives (Pandey, 2024; Ye et al., 2024). The form of

the scaling law used in this paper is consistent with Hoffmann et al. (2022), $L = E + \frac{A}{S^\alpha} + \frac{B}{T^\beta}$, where $\{E, A, B, \alpha, \beta\}$ are fitting parameters. However, we express in a simpler and more appropriate way for our demonstrations.

6.3 Data Mixture Scaling Law

Several studies have examined the scaling laws associated with various data mixture ratios. For instance, Ye et al. (2024) investigate how different data mixtures influence scaling laws during the pre-training phase. However, their proposed laws are not applicable to CPT. Another study by Que et al. (2024) aims to identify the optimal data mixture ratio using the D-CPT law. Their method focuses solely on minimizing domain loss by fixing model sizes and training token volume, thereby neglecting the trade-off between general loss and domain loss, which is critical in CPT.

7 Conclusion

In this work, we investigated the scaling behavior of LLMs under Continual Pre-Training (CPT) to address the limitations of domain-specific performance. We provided a clear definition of Critical Mixture Ratio (CMR) for optimizing the mixture ratio of general and domain-specific data. Our experiments revealed a power-law relationship between loss, mixture ratio, and training data size, allowing us to predict the CMR efficiently. These findings offer practical guidelines for optimizing LLM training, ensuring both general and domain-specific performance while minimizing resource consumption. Furthermore, our study highlights the importance of understanding CPT process and scaling laws, paving the way for future research in this area to enhance LLM capabilities in specialized fields.

8 Limitations

Computational Constraints We experimented with model sizes range from 400M to 3.1B. However, the largest model in our experiments is still relatively small among contemporary LLMs. It may lead to inaccuracy in estimation of model size scaling.

Limited Domains In this work, we conducted continual pre-training only on two specific domains (finance and academic papers) respectively. Although we have draw some useful conclusions from the experimental results, experiments with more domains are expected to provide more refined results and likely to bring some new insights.

CMR scaling law with model size The CMR scaling in this work can only predict the CMR of a fixed model size. We have not explored how to predict CMR of large models with experiments on small models. A possible method is that first we extrapolate all the losses of small models to large models with model size scaling law, and use CMR scaling law to predict the CMR of the large model. We left it as a future work to predict CMR by leveraging multiple scaling laws with less computational efforts.

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714	Martinet, Marie-Anne Lachaux, Timothée Lacroix,	B.1 Notation	766
715	Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal	Summary of notation appearing in the paper:	767
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Table 1: Configurations of the LLMs.

Model Size	460m	940M	1.6B	3.1B
hidden size	1024	1536	2048	2560
intermediate size	3072	4608	6144	7680
number of attention heads	32	32	32	32
number of layers	24	24	24	32
vocabulary size	65632	65632	65632	65632

- F - the loss function of CPT expressed as the Lagrangian.

- T - the amount of training tokens (related to number of iterations, training steps, or the total volume of training data).

- λ - the Lagrange multiplier used to enforce the constraint on the general loss while minimizing the domain-specific loss.

- $T_{\max}$ - the maximum training tokens for CPT.

- T_0 - a point on the training curve where, after training at T_0 and continuing the training, the feasible mixture ratio is observed.

- $\mathbb{F}$ - the set of Feasible Mixture Ratios (feasible mixture ratio).

- R^* - the Critical Mixture Ratio (CMR), which is the optimal mixture ratio that minimizes the loss function within the feasible set.

- $\alpha_1, \alpha_2, \alpha_3$ - parameters to be fitted representing coefficients in the power-law functions for the increment of loss.

- β_1, β_2 - parameters to be fitted representing constants in the power-law functions for the increment of loss.

- s_1, s_2, s_3 - parameters to be fitted representing the exponents in the power-law functions for the increment of loss.

B.2 Feasible mixture ratio

Given that N is fixed, the Objective of CPT in Equation 3 can be transformed into:

$$F(R, T, \lambda) = (\mathcal{L}_{\text{dom}}(M_S) + \mathcal{L}_{\Delta\text{dom}}(R, T)) + \lambda(\mathcal{L}_{\Delta\text{gen}}(\mathcal{R}, T) - \epsilon), \quad (9)$$

where $\mathcal{L}_{\text{dom}}(\text{CPT}(M_S; \mathcal{D}_R, T))$ is split into the value at $T = 0$, $\mathcal{L}_{\text{dom}}(M_S)$ and the increment $\mathcal{L}_{\Delta\text{dom}}$. The corresponding

$$R^* = \underset{R}{\text{argmin}} F(R, T, \lambda) \quad \text{s.t.} \quad \begin{cases} \mathcal{L}_{\Delta\text{gen}}(R, T) \leq \epsilon \\ R \geq 0 \\ T_{\max} \geq T \geq 0 \\ \lambda \geq 0. \end{cases} \quad (10)$$

For a given mixture ratio R , if the training progresses (T increases), and the objective function (Equation 9) shows a decreasing trend, it indicates that the current proportion can lead to the continuation of training towards the expected goal. The trend of the objective function F increasing with training can be reflected by its partial derivative with respect to T :

$$\begin{aligned} & \left. \frac{\partial F(R, T, \lambda)}{\partial T} \right|_{R, \lambda} \\ &= \left. \frac{\partial (\mathcal{L}_{\text{dom}}(M_S) + \mathcal{L}_{\Delta\text{dom}}(R, T))}{\partial T} \right|_{R, \lambda} + \lambda \left. \frac{\partial (\mathcal{L}_{\Delta\text{gen}}(R, T) - \epsilon)}{\partial T} \right|_{R, \lambda}. \end{aligned} \quad (11)$$

Since $\mathcal{L}_{\text{dom}}(M_S)$ and $-\lambda\epsilon$ are constants with respect to T , their derivatives are zero. Thus, we simplify to:

$$\begin{aligned} \left. \frac{\partial F(R, T, \lambda)}{\partial T} \right|_{R, \lambda} &= \left. \frac{\partial \mathcal{L}_{\Delta\text{dom}}(R, T)}{\partial T} \right|_{R, \lambda} \\ &+ \lambda \left. \frac{\partial \mathcal{L}_{\Delta\text{gen}}(R, T)}{\partial T} \right|_{R, \lambda}. \end{aligned} \quad (12)$$

If the training objective under the fixed ratio progresses as expected, there should be at least one point during the training process ($0 \leq T \leq T_{\max}$) where this partial derivative is less than or equal to 0. From this, we can define a feasible proportion curve that should satisfy the following inequality conditions:

$$\exists T \in [0, T_{\max}] : \left. \frac{\partial F(R, T, \lambda)}{\partial T} \right|_{R, \lambda} \leq 0 \quad (13)$$

This means that we only need to determine whether the solution T of the above inequality 13 belongs to $[0, T_{\max}]$ in order to judge whether the current training meets the target. Setting it equal to zero to figure out:

$$\left. \frac{\partial F(R, T, \lambda)}{\partial T} \right|_{R, \lambda} = 0 \quad (14)$$

Setting the equation to zero and further simplifying to express it :

$$\left. \frac{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)}{\partial T} \right|_R + \lambda \left. \frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial T} \right|_R = 0 \quad (15)$$

To derive the following equation using the chain rule:

$$\left. \frac{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)}{\partial T} \right|_R = -\lambda \left. \frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial T} \right|_R \quad (16)$$

By isolating $\left. \frac{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)}{\partial T} \right|_R$, we get:

$$\left. \frac{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)}{\partial T} \right|_R = -\lambda \left. \frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial T} \right|_R \quad (17)$$

Using the chain rule, we have:

$$\left. \frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial T} \right|_R = \frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)} \left. \frac{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)}{\partial T} \right|_R \quad (18)$$

By substituting this into the given equation, we get:

$$\left. \frac{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)}{\partial T} \right|_R = -\lambda \left(\frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)} \cdot \left. \frac{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)}{\partial T} \right|_R \right) \quad (19)$$

Assuming $\left. \frac{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)}{\partial T} \right|_R \neq 0$, we can cancel the terms:

$$1 = -\lambda \cdot \frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)} \Big|_R \quad (20)$$

Thus, we obtain:

$$\frac{\partial \mathcal{L}_{\Delta \text{gen}}(R, T)}{\partial \mathcal{L}_{\Delta \text{dom}}(R, T)} \Big|_R = -\frac{1}{\lambda} \quad (21)$$

Since $\lambda > 0$, the above derivative is a negative number. For a specific R , if there exist points on the training curve where the partial derivatives of the two Δ values are equal to $\frac{1}{\lambda}$, then the ratio is consistent with the expected goal of continual pre-training. These ratios are called feasible mixture ratios, and their set is denoted as $\mathbb{F}$. This is consistent with the feasible mixture ratios marked in Figure 1.

B.3 Fitting

Following the previous work (Kaplan et al., 2020; Hoffmann et al., 2022), we have adopted the power-law as the parametric forms, which is different from other mixture law study (Ye et al., 2024). Previous work has shown that the model parameter N and the amount of data training T are independently related to the power law of loss. However, one point that our work related to power law is different. First, the function we choose to fit is the increment of Loss. Second, due to the phenomenon of general loss increasing first and then decreasing, in order to better fit the data, we used a two-term power-law function. According to Equation 9, the data mixture scaling law for CPT training is defined as follows:

Given:

$$\begin{cases} \mathcal{L}_{\Delta \text{dom}}(T) = \alpha_1 \cdot T^{s1} + \beta_1, \\ \mathcal{L}_{\Delta \text{gen}}(T) = \alpha_2 \cdot T^{s2} + \alpha_3 \cdot T^{s3} + \beta_2. \end{cases} \quad (22)$$

where $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, s1, s2$, and $s3$ are parameters to be fitted.

First, according to the definition of feasible mixture ratios, we can solve feasible mixture ratios under the setting of data mixture scaling law. As the fitting at this time is an extrapolation of the training quantity, R is a fixed value. For simplicity, we no longer explicitly write R , so both $\mathcal{L}_{\Delta \text{dom}}$ and $\mathcal{L}_{\Delta \text{gen}}$ are univariate functions of T . First, differentiate $\mathcal{L}_{\Delta \text{dom}}(T)$ with respect to T :

$$\begin{aligned} \frac{d}{dT} \mathcal{L}_{\Delta \text{dom}}(T) &= \frac{d}{dT} (\alpha_1 \cdot T^{s1} + \beta_1) \\ &= \alpha_1 \cdot s1 \cdot T^{s1-1}. \end{aligned} \quad (23)$$

Next, differentiate $\mathcal{L}_{\Delta \text{gen}}(T)$ with respect to T :

$$\begin{aligned} \frac{d}{dT} \mathcal{L}_{\Delta \text{gen}}(T) &= \frac{d}{dT} (\alpha_2 \cdot T^{s2} + \alpha_3 \cdot T^{s3} + \beta_2) \\ &= \alpha_2 \cdot s2 \cdot T^{s2-1} + \alpha_3 \cdot s3 \cdot T^{s3-1}. \end{aligned} \quad (24)$$

According to the the expected CPT trend in Equation 13, we need to figure whether the critical T_0 that meets this condition is in the effective range $[0, T_{\text{max}}]$. Therefore, the solution for the Equation 15 is important, which can be solved as Equation 15:

$$\frac{d}{dT} \mathcal{L}_{\Delta \text{dom}}(T) + \lambda \frac{d}{dT} \mathcal{L}_{\Delta \text{gen}}(T) = 0 \quad (25)$$

Substitute Equation 23 and Equation 24 respectively, we get:

$$\begin{aligned} \alpha_1 \cdot s1 \cdot T^{s1-1} &+ \lambda (\alpha_2 \cdot s2 \cdot T^{s2-1} \\ &+ \alpha_3 \cdot s3 \cdot T^{s3-1}) = 0 \end{aligned} \quad (26)$$

Further simplifying:

$$\alpha_1 \cdot s1 \cdot T^{s1-1} + \lambda\alpha_2 \cdot s2 \cdot T^{s2-1} + \lambda\alpha_3 \cdot s3 \cdot T^{s3-1} = 0 \quad (27)$$

To solve for T , we can factor out T terms:

$$T^{s1-1}(\alpha_1 \cdot s1 + \lambda\alpha_2 \cdot s2 \cdot T^{s2-s1} + \lambda\alpha_3 \cdot s3 \cdot T^{s3-s1}) = 0 \quad (28)$$

Therefore, the critical points T_0 can be solved by:

$$T_0^{s2-s1} = -\frac{\alpha_1 \cdot s1}{\lambda\alpha_2 \cdot s2} - \frac{\lambda\alpha_3 \cdot s3 \cdot T_0^{s3-s1}}{\lambda\alpha_2 \cdot s2} \quad (29)$$

Solving for T_0 :

$$T_0^{s2-s1} = -\frac{\alpha_1 \cdot s1 + \lambda\alpha_3 \cdot s3 \cdot T_0^{s3-s1}}{\lambda\alpha_2 \cdot s2} \quad (30)$$

$$T_0^{s2-s1} = -\frac{\alpha_1 \cdot s1}{\lambda\alpha_2 \cdot s2} - \frac{\lambda\alpha_3 \cdot s3 \cdot T_0^{s3-s1}}{\lambda\alpha_2 \cdot s2} \quad (31)$$

Thus, the solution for T_0 in terms of the original parameters is:

$$T_0 = \left[-\frac{\alpha_1 \cdot s1}{\lambda\alpha_2 \cdot s2} \left(1 + \frac{\alpha_3 \cdot s3}{\alpha_2 \cdot s2} T_0^{s3-s2} \right)^{-1} \right]^{\frac{1}{s2-s1}} \quad (32)$$

C Table

Ratio	460m	940M	1.6B	3.1B
100%	1.4628	1.3723	1.3242	1.2585
75%	1.4844	1.3910	1.3416	1.2750
50%	1.5122	1.4155	1.3643	1.2965
33%	1.5387	1.4385	1.3854	1.3170
25%-gt	1.5561	1.4538	1.3994	1.3305
25%-pred	1.5566	1.4546	1.3999	1.3303
Difference	0.03%	0.05%	0.03%	0.02%

Table 2: Domain Proportion and Predicted/Actual Value Relative Error

D Figure

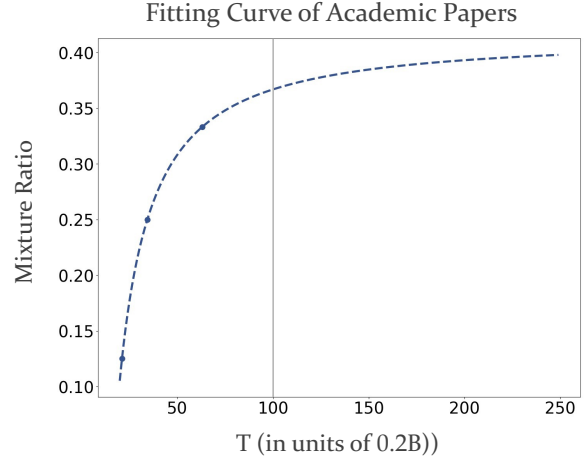


Figure 7: With a fixed model size $S = 460m$, using the CMR scaling law can be extrapolated to $T = 250$ and more. We can use the CMR scaling laws to predict CMR for *Academic Papers* in the CPT of M_{460m} . When $T = T_{\max} = 100$, the value of R is 36.69%, regarded as the CMR.

Metric Ratio		General				Domain			
		460m	940M	1.6B	3.1B	460m	940M	1.6B	3.1B
MSE	100%	1.9394e-10	2.5695e-10	9.8058e-10	2.2880e-11	9.7830e-08	7.6174e-08	6.4577e-08	4.4057e-08
	75%	5.2270e-11	7.7104e-15	3.4402e-12	1.5432e-11	1.2283e-07	7.6940e-08	7.1749e-08	4.4160e-08
	50%	1.5340e-10	4.4162e-11	1.5992e-09	1.6405e-10	1.2539e-07	7.0535e-08	5.1893e-08	3.8559e-08
	33.3%	5.2538e-11	1.1070e-10	5.5883e-11	5.4041e-11	1.1904e-07	6.9371e-08	5.8162e-08	4.4630e-08
	25%	7.3045e-11	4.7677e-11	8.7140e-11	1.6598e-14	1.0966e-07	7.0327e-08	4.7272e-08	4.6702e-08
	12.5%	6.9011e-11	8.9891e-11	7.1858e-11	9.2656e-11	8.1609e-08	7.2597e-08	5.1854e-08	4.4091e-08
R ₂	100%	0.9999	0.9999	0.9998	0.9989	0.9957	0.9969	0.9969	0.9978
	75%	0.9993	0.9999	0.9990	0.9963	0.9954	0.9966	0.9966	0.9975
	50%	0.9973	0.9966	0.9946	0.9593	0.9951	0.9963	0.9965	0.9971
	33.3%	0.9928	0.9877	0.9818	0.9251	0.9954	0.9959	0.9965	0.9967
	25%	0.9872	0.9763	0.9659	0.8741	0.9956	0.9959	0.9966	0.9966
	12.5%	0.9590	0.9438	0.9520	0.8972	0.9974	0.9962	0.9963	0.9965

Table 3: The MSE and R₂ of the fitting power-law of Δ_{General} (Δ_{Domain}) loss by training tokens T

Domain-General Loss Curve (Scale cross-section and Ratio cross-section)

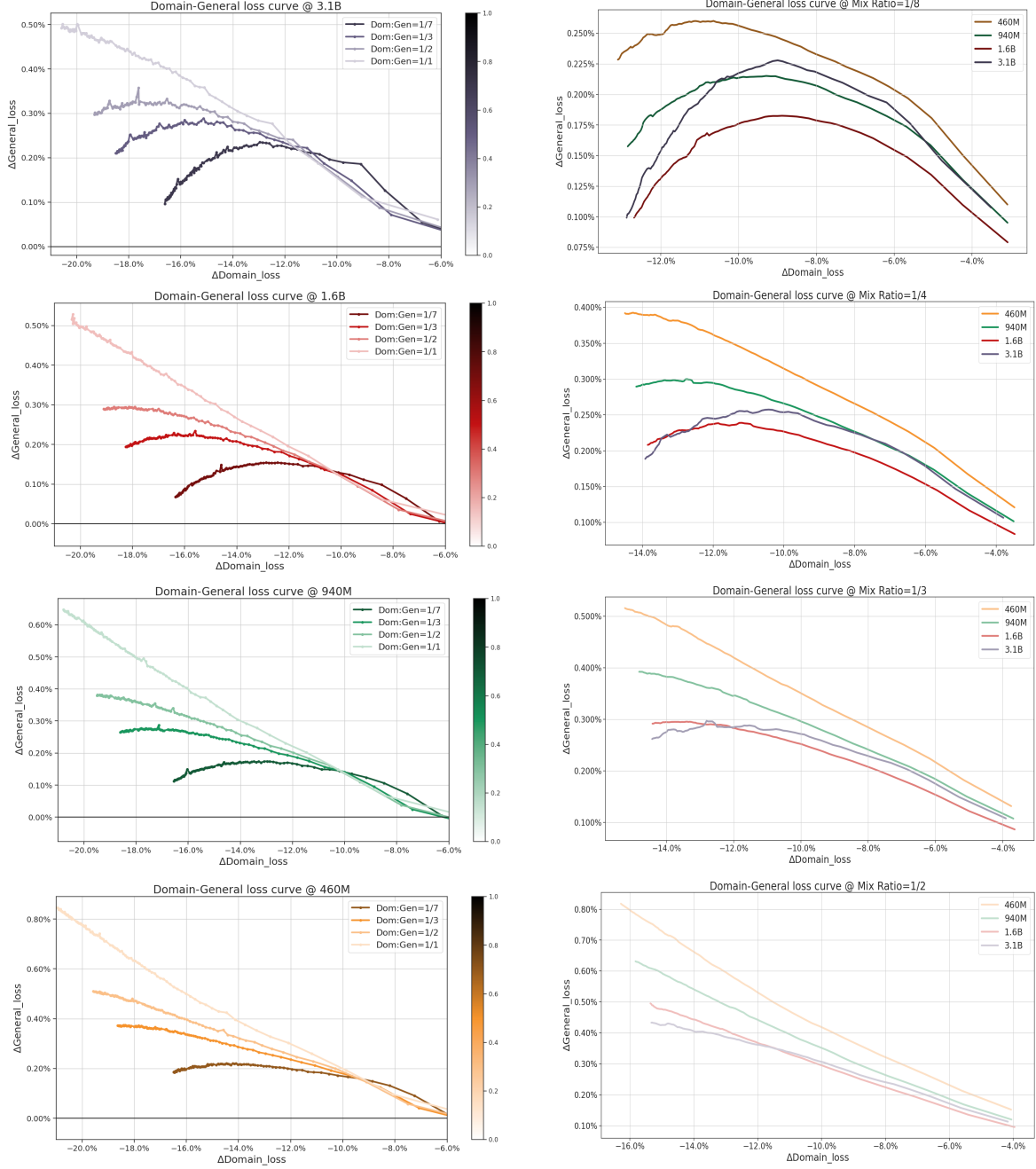


Figure 8: Each cluster represents a different mixing ratio, which is 1/8, 1/4, 1/3, 1/2. Pay attention to the third set of lines, that is, clusters with a proportion of 1/3. The cross-section of this set of lines is shown on the right.

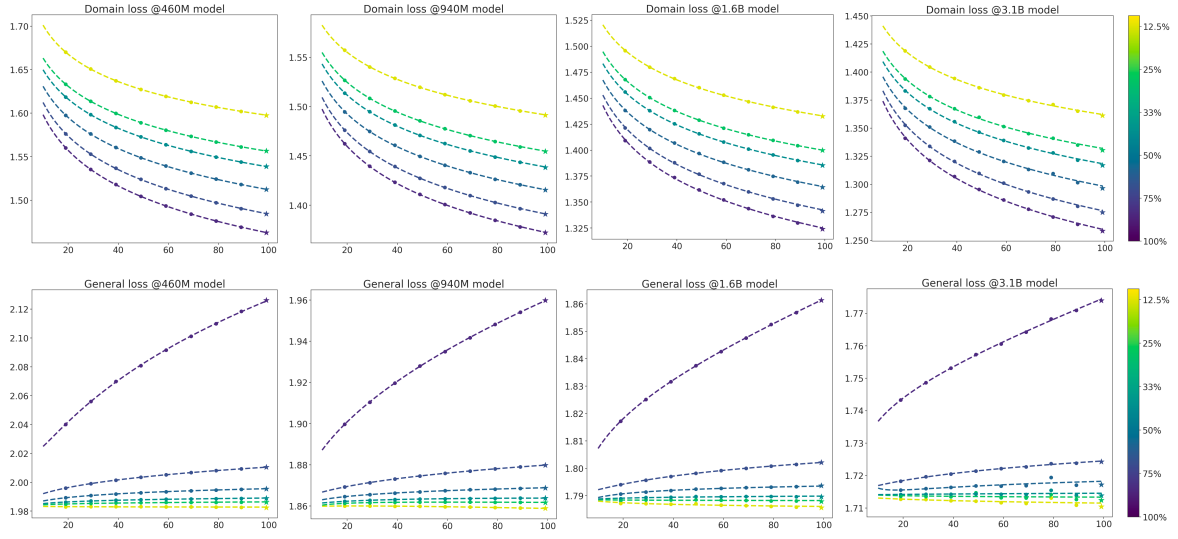


Figure 9: The temperature bar represents the mixture ratio R , which takes six values ranging from $1/8$ to 1 . Different subgraphs are fitting curves that change with the increase of T in the training process for different M_N domain loss and general loss. Overall, the domain loss keeps decreasing during the training process while the general loss keeps increasing. It is worth noting that although the general loss is increasing, the magnitude of its increase is actually very small, especially when the mixture ratio is not very big ($R = \{1/8, 1/4, 1/3, 1/2, 3/4\}$), with a total increase of less than 0.02 . The solid circles (●) represent real losses, and the stars (★) represents the predicted losses. We can see that whether it is general loss or domain loss, the predicted values fall on the fitted curves accurately.

Extrapolation of T with power law

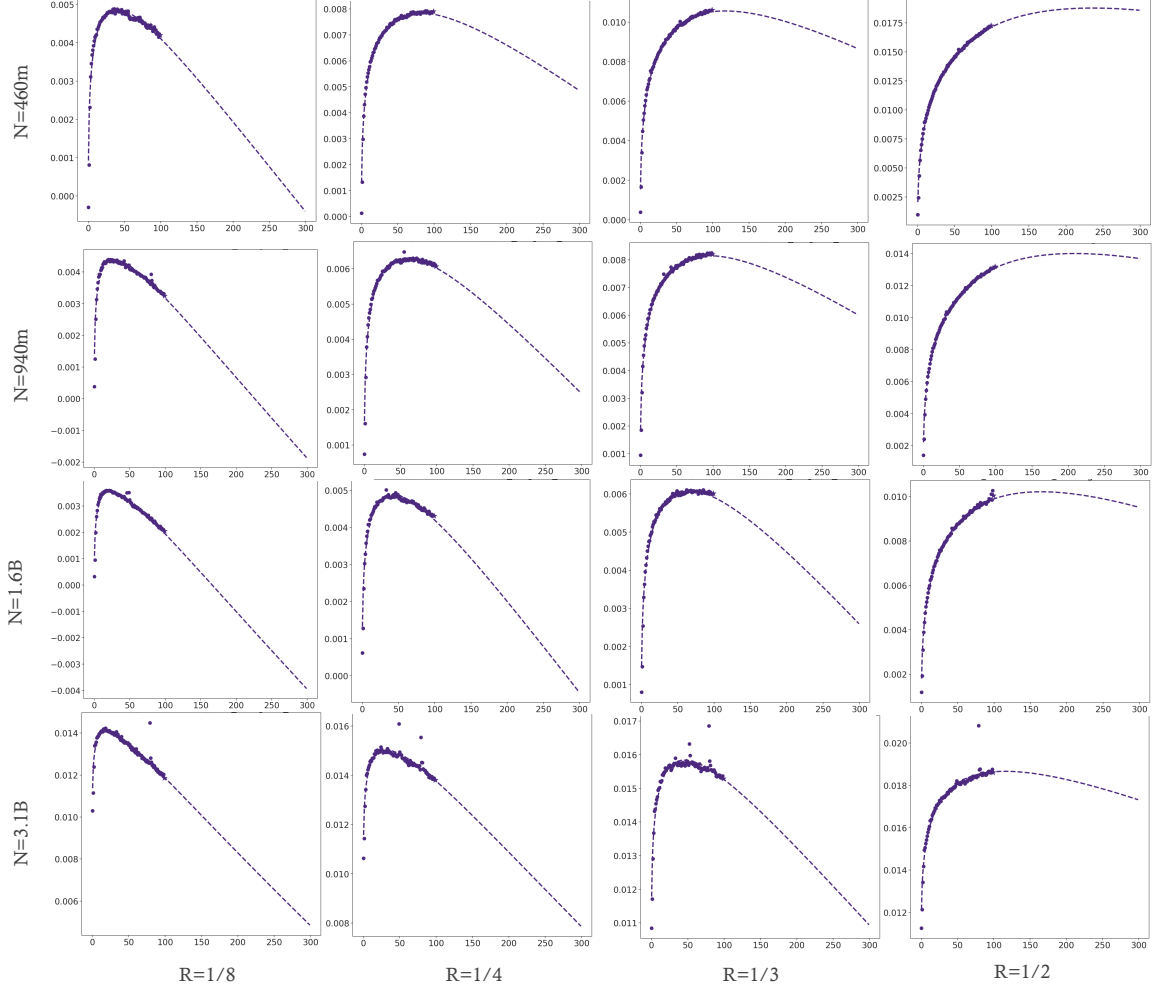


Figure 10: Power laws of training token volume for different model sizes in *Finance*. Compared with the extrapolation of the training volume of the model of the same size to continue training in the *Academic papers* field in Figure 6, it can be seen that under the same proportion, the amount of training volume of CPT of Academic Papers is larger where the inflection point appears.